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Riemannian Manifold Hamiltonian Monte Carlo

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Plan

1. Context
2. Experiments
3. Conclusion

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HMC and RHMC : a better exploration!

Geodesic flows

- Hamiltonian equations : modeling motion dynamics
- Hamiltonian = kinetic energy + potential energy
- RHMC: generalization with non-constant curvature

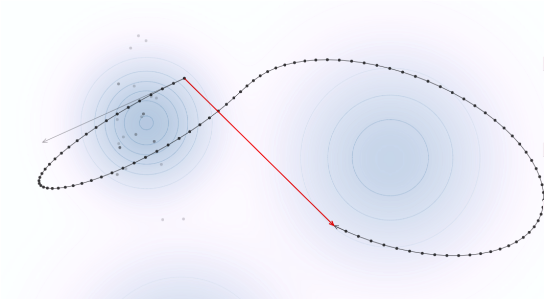

FIGURE 1: HMC from Chi Feng demo

Questions raised by the document studied

Girolami and Calderhead's paper areas of improvements [Girolami2011RiemannML]

- A better integration scheme ?
- A better metric ?
- A numerical estimation of the curvature?

Theorem: two theoretical results

- Verlet's leapfrog is sympletic and assures time reversibility which implies detailed balance [**bishop2006pattern**]
- The geodesic flow for negatively curved compact Riemannian manifolds is ergodic [**Anosov1969-xs**]

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Three improvements already made

Three interesting papers

- Autodifferentiation for Bayesian neural network[**cobb2020scaling**]
- No-U-Turn (NUTS) : The No-U-Turn Sampler [**hoffman2011nouturn**]
- Softabs metrics[**Betancourt˙2013**] : " maintain the desirable behavior of the Hessian in convex neighborhood"

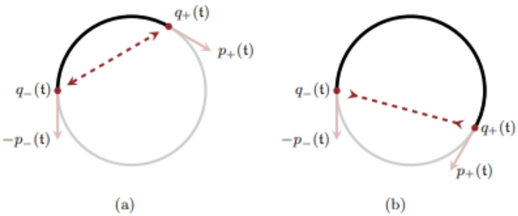

FIGURE 2: NUTS from 'A Conceptual Introduction to Hamiltonian Monte Carlo' [**betancourt2018conceptual**]

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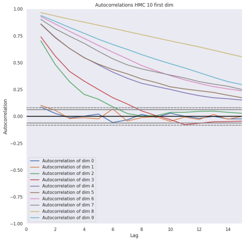
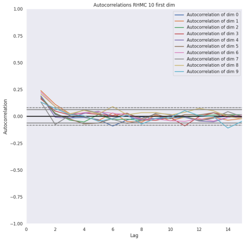
Automatic differentiation and HMC vs RHMC

A trade-off between speed and precision

- Autodifferentiation makes RHMC slower and needs explicit solutions (x50 to x1000).
- Autocorrelations are far better for RHMC (no small oscillations).
- RHMC sampling has smaller KL divergence ($0.3 < 1670$)

HMC vs RHMC 200D Gaussian distribution



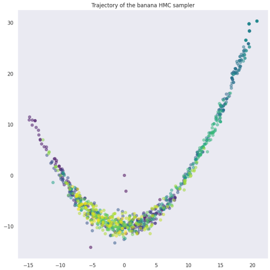


Softabs

More stable but needs better implementation

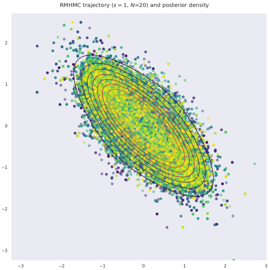
- Too slow to be used with implicite integration.
- Assure SDP stability.
- Should use a specialised framework, STAN ?

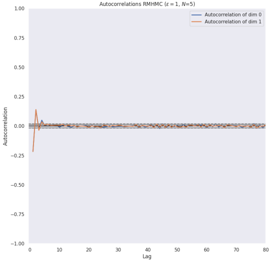
HMC vs RHMC Softabs

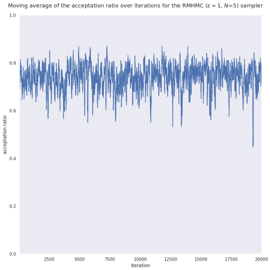


Hyperparameter influences

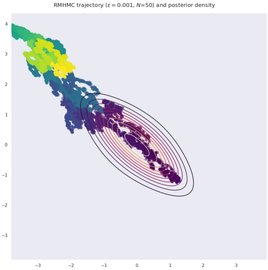
$\varepsilon = 1,$
$N = 5$

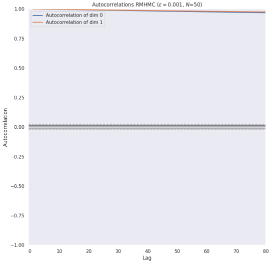


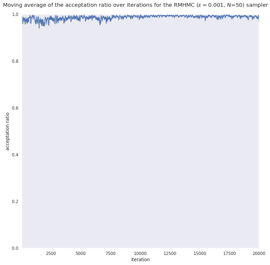




$\varepsilon = 10^{-3},$
$N = 50$







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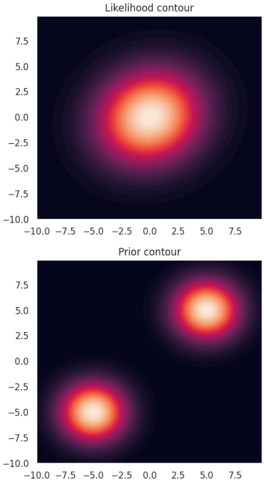
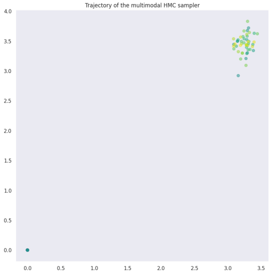
Multimodal distributions for RHMC

Gaussian mixture (RHMC)



HMC



Adding the likelihood to the potential

- $\mathcal{L}(\theta) \to \mathcal{L}(\theta, Y) = \mathcal{L}(\theta) + \mathcal{L}(Y|\theta)$

- Does not help explore other modes in this case ...

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Conclusion & extensions

- Easily implemented and can work really well, but hyperparameter tuning is not simple, done manually in the paper

- Derivation of the (expected) Rao metric is intractable in some cases (e.g. mixture models)
  - Approximation must be used or other metrics chosen
  - The problem of the right choice of matrix is not entirely solved although it gives very good directives

- Could be coupled with parallel tempering methods to better handle the multimodal case

- Alternative geometries could be investigated, for instance with MMD [**briol2019statistical**] or Wasserstein [**li2020wasserstein**] distances

References

The curse of dimensionality

Measure concentration

- Motivates Bayesian inference
  - Integration instead of Optimization
  - Monte Carlo estimation
- Motivates manifold hypothesis

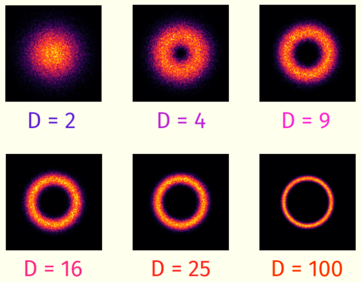


FIGURE 3: Gaussian sampling projections from Jean Feydy course

NUTS HMC using STAN

STAN a hard but powerful framework

- Good performance but bad documentation
- Autograd and RHMC are not upgraded in python yet.
- Needs to create the cython script by yourself for now.

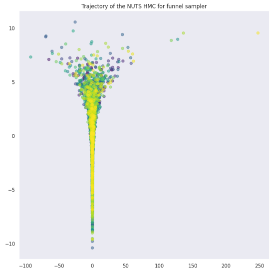


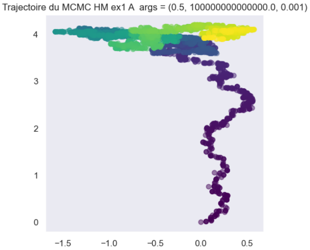
FIGURE 4: NUTS HMC for funnel's distribution

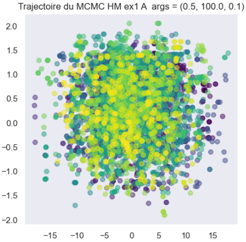
What is an efficient sampling ?

Sampling techniques

- Use of dependencies : MCMC
- Exploration-exploitation dilemma
  - Exploration to have all the "typical" states
  - Exploitation to obtain the right proportions

Exploration Exploitation Metropolis Hastings TP4





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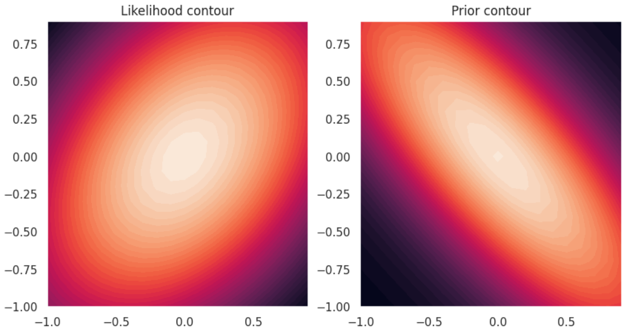
Toy problem

- $Y|\theta \sim \mathcal{N}\left(\frac{1}{2}\theta^{\odot 2}, \Sigma_0\right)$ ($\odot$: Hadamard product)
- Prior $\theta \sim \mathcal{N}(0, \Sigma)$
- Fisher-Rao metric tensor $G(\theta) = \mathrm{diag}(\theta)\Sigma_0^{-1}\mathrm{diag}(\theta) + \Sigma^{-1}$



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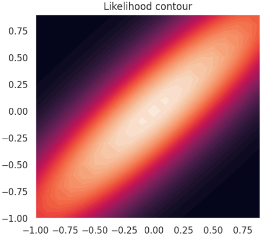
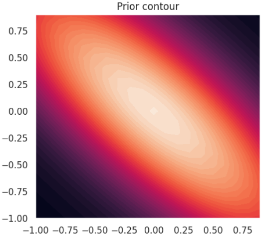
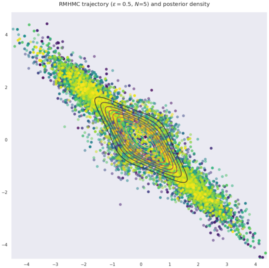
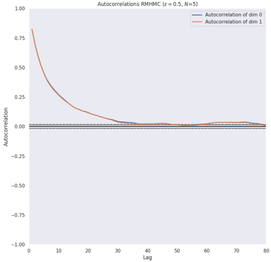
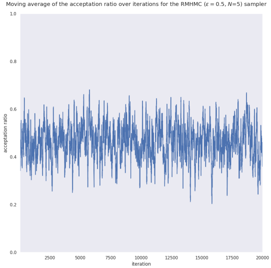
Experiments using our own implementation











Multimodal distributions for HMC

HMC

- HMC can be computed but doesn't explore enough.

HMC 2D Gaussian Mixture and Banana-shape